

P2H2P wellbore storage

A primer: how the model works

Companion to `P2H2P_model.ipynb`, model version 0.6

Craft & Hawkins Department of Petroleum Engineering
Louisiana State University

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Abstract

This is a companion to the Colab notebook `P2H2P_model.ipynb`: the same material, with room to explain why each piece is the way it is. It is not the manuscript. Nothing here is a defence of the model against a reviewer — there are no verification results, no comparisons with superseded formulations and no version history. Those live in `P2H2P_verification.ipynb` and in `DESIGN_NOTES.md`, and you should not need them to use or to modify the model.

Read §1 and §2 first; they are the shape of the whole calculation. Sections 4–8 are the physics, in the order the code evaluates it. Section 10 is the one to keep beside you when you start changing things.

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1 What the plant does, and what the model computes

Electricity is stored as heat and returned as electricity. During charging a high-temperature heat pump raises heat from a warm source and delivers it to pressurised water, which is pumped down

a repurposed oil-and-gas well through finned hairpin tubes. The annulus around those tubes is packed with a *phase-change material* (PCM), which melts and stores the energy as latent heat. During discharging the flow reverses, the PCM freezes, and the recovered heat drives an organic Rankine cycle (ORC) that returns electricity to the grid. The intended duty is 10 h charging and 10 h discharging, with two 2 h idle periods, repeated daily.

The engineering question is how many wells a field needs to deliver a specified electrical output — 1 MW here — and what limits it.

Everything turns on one difficulty. The PCM conducts heat poorly: for the material used here $k \approx 0.5 \text{ W/m/K}$, a hundred times worse than steel. As the melt layer grows, the heat has further to travel through a bad conductor to reach the remaining solid, so the exchanger gets worse as it fills. A useful model must therefore couple a *heat transfer network* — convection, tube wall, fins, and that growing melt layer — to a *moving phase boundary*, over a duty cycle, cheaply enough to run hundreds of times.

The cascade. One feature of this design shapes everything downstream. The water enters hot and leaves cold: it *glides* by $\Delta T_{3C,2C} = 55 \text{ K}$ along the well. A single PCM would therefore be barely melted by the cold end of the water and strongly superheated by the hot end. So the PCM is divided into N_{lay} **layers of different melting temperature**, arranged so that hot water meets high-melting material and cold water meets low-melting material. Each layer sees a workable temperature difference. This is why almost every profile in this document has a staircase in it.

2 The calculation, end to end

Figure 1 is the whole thing. It is worth reading before any of the physics, because the structure is slightly unusual: **nothing is solved for**. The thermodynamic cycles, the energy budget and the exchanger size are evaluated in sequence; the two mass flow rates follow from prescribed temperature glides; and the only iteration anywhere is the outer loop that repeats the daily cycle until the store returns to its own initial state. Section 8 explains why it *has* to be arranged that way.

Three levels are nested, and they are almost independent:

- the **plant** level — heat pump and ORC — is evaluated once, and passes only two efficiencies and a handful of water temperatures downstream;
- the **field** level turns the 1 MW target into an energy per cycle, and hence a well count;
- the **borehole** level marches one exchanger through a half-cycle, and is where all the difficulty lives.

The cost is dominated by the innermost pair of loops: n_t time levels $\times$ n_s segments $\times$ two half-cycles $\times$ the number of cycles to convergence — about 2×10^5 segment updates for the default case, which takes a few tens of seconds.

3 What is assumed

Every one of these is a simplification you could relax, and three of them are worth watching in a parametric study.

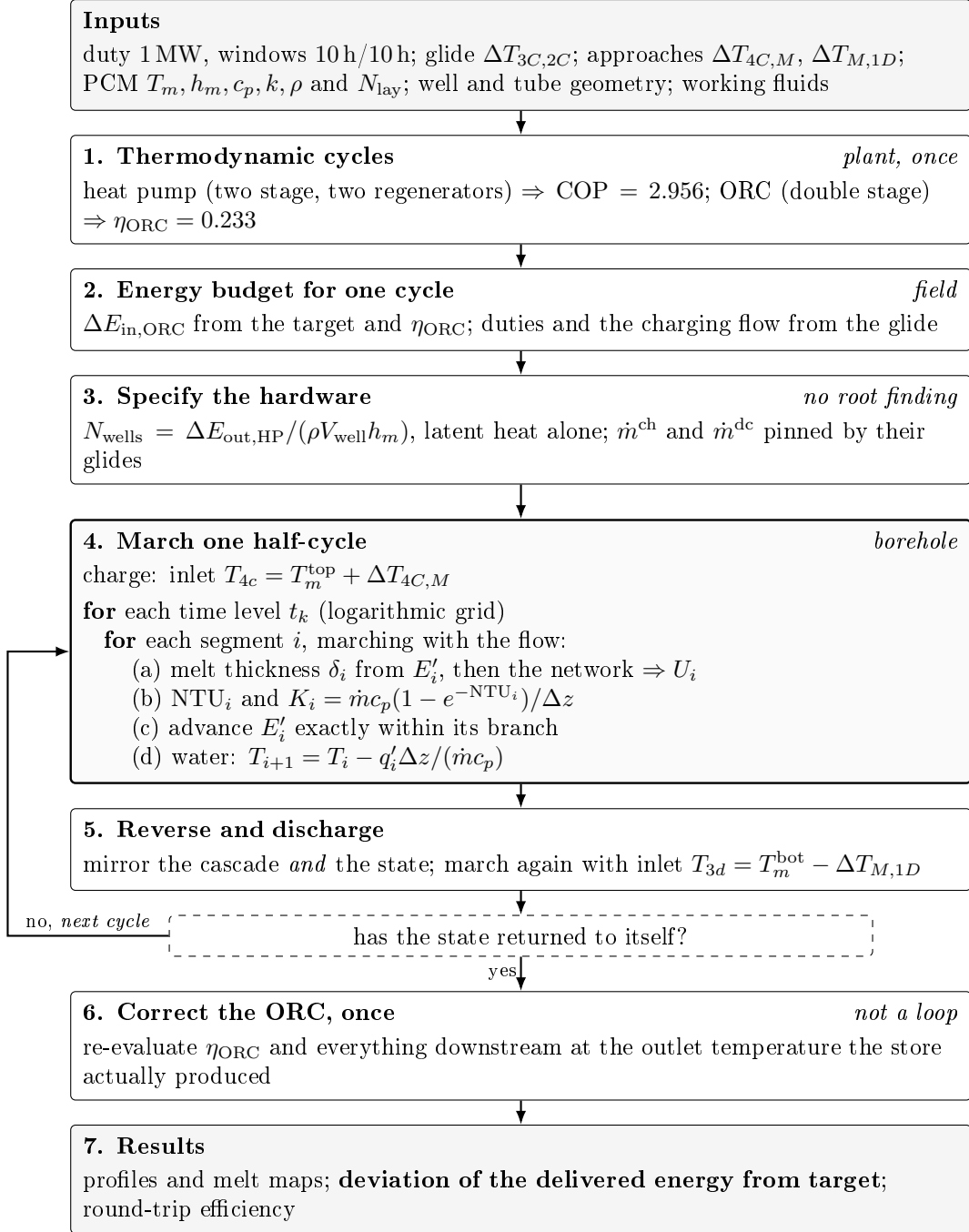


Figure 1: The procedure. Steps 1–3 are evaluated once and produce a fully specified piece of hardware; steps 4 and 5 are the exchanger model, run twice per cycle; the loop back from the test is the only iteration in the calculation.

	assumption	where it bites
H1	the PCM in a segment is one lumped state	no radial temperature profile inside the melt
H2	no axial conduction between segments	the idle periods do nothing at all
H3	the melt region is a concentric annulus	saturated at the design point: neighbouring fronts touch over 99 % of the well
H5	one melt thickness per segment	see H1
H6	the borehole wall is adiabatic	no ground loss, so the store is lossless by construction
H7	the water is quasi-steady	the first 38 min of each half-cycle is not resolved
H8	the cascade is laid along the developed tube length	the two legs of a hairpin carry different T_m at the same depth
H9	no heat exchange between the two legs	they are up to 48.9 K apart at the wellhead

H7 deserves a sentence, because it is what makes the model cheap. The water crosses the whole tube in 2264 s while the PCM state evolves over ten hours — a separation of about sixteen — so the water profile is treated as instantaneous at each time level. The price is that the startup transient of each half-cycle is not resolved. What makes that tolerable is that the energy held in the water is only 1.2 % of the latent inventory it is charging, so the unresolved transient misplaces very little energy even though it occupies a noticeable fraction of the window.

4 The state variable, and why it is enthalpy

Divide the developed tube length into n_s segments. Each segment owns a slice of PCM, and the model carries **one number** for it.

The obvious choice is the melt thickness δ , or equivalently the melted cross-sectional area A_{melt} — it is what you would draw. It fails, for a reason worth understanding before you trust anything else in the model.

Ask such a model for the temperature of the PCM in a segment. The only available answer is T_m : that is all the state encodes. While solid remains, that is correct, which is why the limitation is easy to miss. It stops being correct the moment a segment finishes melting. The network still reports a positive heat flow — the water is still hotter than T_m , and nothing in the resistance chain knows the PCM is exhausted — and the model must do one of two things. It can keep growing A_{melt} , melting material that is not there and producing melted fractions above unity. Or it can clip A_{melt} at the available inventory and throw the surplus away, in which case the melted fraction is physical but energy is not conserved and the water leaves at a temperature that reflects heat which went nowhere.

Neither is acceptable, and no refinement of the marching scheme repairs either, because the deficiency is in the *state*: a variable counting melted area has no slot for liquid above T_m or solid below it.

The fix. Carry the **enthalpy** E' per unit tube length, measured from a datum of fully solid material at that segment's own melting temperature. With A_{avail} the PCM cross-section belonging to one tube, define

$$E'_{\text{lat}} = \rho h_m A_{\text{avail}}, \quad C_s = \rho c_{p,s} A_{\text{avail}}, \quad C_l = \rho c_{p,l} A_{\text{avail}}, \quad (1)$$

the latent capacity and the two sensible capacities. The PCM temperature and the melted area then follow from E' alone:

$$T_{\text{pcm}} = \begin{cases} T_m + E'/C_s, & E' < 0 \quad (\text{subcooled solid}), \\ T_m, & 0 \leq E' \leq E'_{\text{lat}} \quad (\text{two-phase}), \\ T_m + (E' - E'_{\text{lat}})/C_l, & E' > E'_{\text{lat}} \quad (\text{superheated liquid}), \end{cases} \quad (2)$$

$$A_{\text{melt}} = \frac{\min(\max(E', 0), E'_{\text{lat}})}{\rho h_m}, \quad \varepsilon_{\text{local}} = \frac{A_{\text{melt}}}{A_{\text{avail}}} \in [0, 1]. \quad (3)$$

Figure 2 is the reason this works. Read as $E'(T)$ the relationship is multivalued at T_m — which is exactly why temperature cannot be the state variable. Read as $T(E')$, the direction used here, it is single-valued everywhere, and position along the vertical segment *is* the melted fraction.

The saturation in Eq. (3) is not a guard bolted on to suppress bad behaviour; it is the definition. Melted area saturates because a segment cannot melt PCM it does not own, and the extra energy has somewhere to go: superheat. So $\varepsilon_{\text{local}} \in [0, 1]$ is *structural*. No code path can violate it and no test for it is needed.

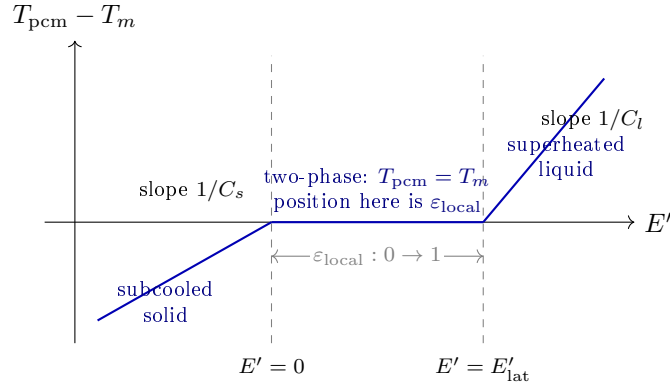


Figure 2: The constitutive curve, Eq. (2). One scalar distinguishes three physical states, and the melted fraction is where you are along the flat part.

How much is at stake. It would be reasonable to guess this is a small correction. One kelvin of superheat is worth $C_l/E'_{\text{lat}} = c_{p,l}/h_m = 0.47\%$ of the latent capacity, so with a nominal 10 K approach the sensible branches look like 4.7% of the energy. The measured figure at the design point is 9.3% — 7.1% as superheat, 2.3% as subcooling — because 10 K is the approach of the *first* cascade layer and a saturated segment goes on superheating past it, reaching 18.5 K here. And the store finishes every charge fully molten in *all* 100 segments, so an area-based model is clipping everywhere at the moment the store is fullest.

That number is worth recomputing whenever you change the PCM: the ratio $c_{p,l}/h_m$ is the relevant group, so halving the latent heat doubles it.

5 The segment equation

Two control volumes, taken together: the PCM belonging to segment i , and the water passing over it.

On the PCM side, all the heat arriving changes the enthalpy, by definition of E' :

$$\frac{dE'_i}{dt} = q'_i, \quad (4)$$

with q'_i the heat per unit tube length crossing into the segment. On the water side, with H7 the water carries no storage, so what it gives up is what the PCM takes:

$$T_{i+1} = T_i - \frac{q'_i \Delta z}{\dot{m}c_p}, \quad \Delta z = \frac{L_{\text{tube}}}{n_s}. \quad (5)$$

Marching Eq. (5) from the inlet gives the water profile along the well.

What closes the pair is the relation between q'_i and the temperatures. It is tempting to write $q' = U_i P_i (T_f - T_{\text{pcm}})$ with U_i the overall coefficient and P_i the perimeter. **That is wrong**, and in a way that matters.

Consider what $U_i P_i$ says as the wall conductance improves: q' grows without limit. But the water can only give up what it carries, and a segment cannot extract more than $\dot{m}c_p$ times the temperature difference available. Deriving the closure properly from the two balances gives the effectiveness form,

$$q'_i = K_i (T_i - T_{\text{pcm},i}), \quad K_i = \frac{\dot{m}c_p (1 - e^{-\text{NTU}_i})}{\Delta z}, \quad \text{NTU}_i = \frac{2\pi r_i U_i \Delta z}{\dot{m}c_p}. \quad (6)$$

As $\text{NTU} \rightarrow 0$ this reduces to $U_i P_i$; as $\text{NTU} \rightarrow \infty$ it saturates at $\dot{m}c_p / \Delta z$, which is the capacity-rate limit. Over a charge in the default case NTU runs from 0.083 to 0.602, so the two forms differ by 4% to 33% — largest early, when the melt layer is thin and the exchanger is at its best.

Which shell the heat crosses — and why it is not obvious. The PCM-side term in U_i is a conduction shell growing *outward from the tube wall*, $R' = \ln(1 + \delta/r_e)/2\pi k$. So the model has to say what that shell is made of, and the answer is not the same in the two half-cycles.

The front always grows away from the tube, because the tube is the driven boundary in both directions. While **melting**, the shell is liquid and its thickness comes from A_{melt} : heat crosses the melt to reach the solid still waiting beyond it. While **freezing**, solidification also starts at the wall, so the shell is the *frozen* material — thickness from $A_{\text{avail}} - A_{\text{melt}}$, conductivity k_s — and the liquid is pushed outward past the front, where it is no longer in the path at all.

	shell against the tube	thickness from	k
melting	liquid	A_{melt}	k_l
freezing	frozen solid	$A_{\text{avail}} - A_{\text{melt}}$	k_s

Written that way the four limiting states need no special handling: fully solid and melting, or fully melted and freezing, both give $\delta = 0$ and no resistance — nothing has grown yet; the two saturated states give the full annulus with the appropriate conductivity.

This is worth pausing on because **the model got it wrong until v0.7**, and the way it stayed wrong is instructive. The melted thickness was used in *both* directions. That is right for melting, and every result that mattered for a long time was a melting result. For freezing it puts the conduction path on the wrong side of the front — and since the two thicknesses are complementary, the error is a reversal in time rather than a scale factor. A segment that had frozen solid was assigned *zero* resistance, exactly where the physical resistance is largest, so the modelled exchanger improved as it froze. The symptom was an overall coefficient seven times larger at the end of the discharge than at the end of the charge, and that was explained away as solid PCM conducting better than liquid — but $k_s/k_l = 1.33$, and nothing accounted for the other factor of five.

One question to the code would have found it: *what resistance do you give a segment at $\varepsilon_{\text{local}} = 0$ during a discharge?*

And when there is no front at all. That rule answers “which side of the front”, which assumes there is one. A cell that is *entirely* one phase — all subcooled solid, or all superheated liquid — has none. Ask the rule anyway and it gives a resistance that depends on which way the heat is going: zero one way, the whole annulus the other. Same body, same material, two answers. That cannot be right, and until v0.8 it is what the model did.

What changes in a single-phase cell is the *meaning* of T_{pcm} . In two-phase it is the temperature of the **front**, so the right resistance is tube-to-front. In single phase it is the **volume mean**, so the right resistance is mean-to-surface — a different question with a different answer.

Take the cell as an annulus $r_e \leq r \leq r_o$, adiabatic at r_o (the neighbouring cell is identical, so that face is a symmetry plane), storing sensible heat at a uniform volumetric rate s . Quasi-steady, whatever crosses radius r is what the material beyond it stores:

$$-k 2\pi r \frac{dT}{dr} = s \pi (r_o^2 - r^2) \implies T(r_e) - T(r) = \frac{s}{2k} \left[r_o^2 \ln \frac{r}{r_e} - \frac{r^2 - r_e^2}{2} \right].$$

Average over the volume, divide by $Q' = s\pi(r_o^2 - r_e^2)$, and you get $R'_{\text{bulk}} = S/2\pi k$ with $\beta = r_o/r_e$ and

$$S = \frac{\beta^4 \ln \beta - \beta^4/2 + \beta^2/2 - (\beta^2 - 1)^2/4}{(\beta^2 - 1)^2}.$$

S depends on geometry alone — k cancels — so one shape factor does for both phases and only the conductivity changes. Here $\beta = 2.109$ and $S = 0.3467$ against $\ln \beta = 0.7460$ for the surface-to-surface annulus: smaller by 2.152, which is the cylindrical version of the 1/3 you already know for a slab with uniform generation.

It goes back into the network as an equivalent thickness, $\delta_{\text{eq}} = r_e(e^S - 1) = 8.74 \text{ mm}$, so the fin efficiency and finned-area machinery apply unchanged. The two values it replaces — 0 and $\delta_{\text{merge}} = 23.37 \text{ mm}$ — bracket it from either side.

A discontinuity that is real. R' jumps at each branch boundary. As $\varepsilon_{\text{local}} \rightarrow 1^-$ while melting the front is at r_o and $R' = \ln \beta / 2\pi k_l$; the moment the cell becomes superheated, $R' = S/2\pi k_l$. Nothing physical moved — the reference temperature changed from the front to the mean, and the resistance follows it. The same happens at $\varepsilon_{\text{local}} \rightarrow 0$, where the mechanism switches from *warming a body* to *moving a front*. The flux stays bounded because K saturates at the capacity rate.

One consequence is worth knowing before you read any U_i plot: **the exchanger is not at its best at $t = 0$** . It is at its best shortly *after* melting starts, when there is a front at the tube and almost nothing between them:

t [h]	$\varepsilon_{\text{local}}$	U_i	which resistance
0.00	0.000	260.2	bulk, subcooled solid
1.16	0.039	479.4	shell, thin melt — the best moment of the run
3.41	0.245	214.8	shell, thicker
10.00	1.000	213.7	bulk, superheated liquid

What one thickness cannot do. One δ describes one front. At cyclic steady state the charge begins with residual liquid in the *outer* part of the cell, so melting then produces liquid at the tube, solid in the middle and liquid outside — three regions. No single thickness represents that. The arrangement above is the better approximation, not a correct one, and that is what hypothesis H5 really assumes.

Where U_i comes from. U_i is the usual series network referred to the inner tube area: internal convection (Gnielinski), the fouling resistance, conduction through the tube wall, and then the PCM side — the finned outer surface with its fin efficiency, in series with conduction across the annular melt layer of current thickness δ_i . That last term is the one that changes as the segment melts, and it is what makes the problem nonlinear and interesting: the exchanger degrades as it fills. The melt thickness is recovered from A_{melt} by inverting the finned-annulus area relation, which is why `delta_from_area` exists.

6 Integrating in time

Substituting Eq. (6) into Eq. (4) gives, for one segment,

$$\frac{dE'}{dt} = K (T_f - T_{\text{pcm}}(E')). \quad (7)$$

This is nonlinear, because $T_{\text{pcm}}(E')$ is piecewise — but it is *affine within each branch* of Eq. (2). That structure is worth exploiting, for a reason that is not obvious.

On the plateau, $T_{\text{pcm}} = T_m$ is constant, so the right-hand side does not depend on E' at all: the capacity is effectively infinite and explicit Euler is *exact* for any step size. On the sensible branches it is not. There Eq. (7) is a relaxation with time constant C/K , and explicit Euler requires $\Delta t < C/K$.

The time grid here is logarithmic — the natural choice for a diffusion-limited process, where everything happens quickly at first and then slowly. Its last step spans 8491 s against a time constant of 1102 s: a ratio of 7.7. An explicit implementation duly diverges, and it does so spectacularly, alternating in sign and passing $8e10$ K within twelve steps.

The scheme. Rather than refine the grid by three orders of magnitude, integrate each branch in closed form. Holding T_f fixed across the step,

$$E'(t + \Delta t) = \begin{cases} E' + K (T_f - T_m) \Delta t, & \text{plateau,} \\ E'_{\text{eq}} + (E' - E'_{\text{eq}}) e^{-K \Delta t / C}, & \text{sensible branches,} \end{cases} \quad (8)$$

where E'_{eq} is the enthalpy at which the PCM would sit in equilibrium with the water — the state the segment relaxes toward and never reaches. The exponential is unconditionally stable and monotone, so no step, however large, can overshoot it. That is precisely the property explicit Euler lacks.

A step may begin in one branch and end in another, in which case it is split at the crossing: advance to the boundary, then continue in the adjacent branch with the remaining time. Only a few crossings can occur within one step, so the loop is bounded.

One segment, one step.

1. $t_{\text{left}} \leftarrow \Delta t$
2. **while** $t_{\text{left}} > 0$:
3. identify which branch contains E'
4. compute the time t^* to reach that branch's boundary
5. **if** $t^* \geq t_{\text{left}}$: advance by Eq. (8) and stop
6. **else**: move to the boundary, subtract t^* , repeat

Energy closure is then structural rather than checked: whatever the scheme puts into E' is exactly what it took out of the water.

7 The cascade, and which temperatures you may prescribe

The layers are graded so that each sits $\Delta T_{4C,M}$ below the charging water at *its own* leading face, assuming the water glides linearly. Since there are N_{lay} equal layers sharing a glide of $\Delta T_{3C,2C}$, they are spaced one layer width apart and the coldest one is

$$T_m^{\text{bot}} = T_m^{\text{top}} - \Delta T_{3C,2C} \frac{N_{\text{lay}} - 1}{N_{\text{lay}}}. \quad (9)$$

Note the factor $(N_{\text{lay}} - 1)/N_{\text{lay}}$: the span of melting temperatures is *one layer width short* of the glide, because the last layer is referenced to the water entering it rather than leaving the well. For the default case that is 48.889 K of cascade against a 55 K glide.

Only the inlets are boundary conditions. This is the single most important structural point in the model, and it is easy to get wrong. The exchanger has two inlets and two outlets. The *inlets* are prescribed; the *outlets* are computed. You cannot specify both, and attempting to specify an outlet produces a model that contradicts itself.

$$\underbrace{T_{4c} = T_m^{\text{top}} + \Delta T_{4C,M}}_{\text{charge: superheat, drives melting}}, \quad \underbrace{T_{3d} = T_m^{\text{bot}} - \Delta T_{M,1D}}_{\text{discharge: subcooling, drives freezing}} \quad (10)$$

Why it matters: under Eq. (2) the store finishes each charge superheated, so the water leaving at the start of the next discharge is at 157.93 °C — *above* the melting temperature of the hottest layer. Any prescribed value for that outlet is contradicted by the solution itself. The outlet is a result, and it is reported as one.

The discharge runs backwards. The flow reverses between half-cycles, so the discharge march starts at the *cold* end of the cascade. Both the melting temperatures and the melt state must be mirrored with the march index — and the state especially, because E' is measured from a datum of *solid at the local T_m* , so reversing the cascade without reversing the state moves the datum by up to 48.9 K. Results are returned in depth coordinates, so that there is one cascade rather than two.

8 Cyclic steady state, and why sizing is a simulation

Everything so far describes one half-cycle from a given starting state. A plant repeating a daily schedule does not start from a cold, fully solid store after the first day, so the cycle must be repeated until the state reproduces itself. That takes about 70 cycles here, and the count grows with the exchanger count.

At that point something restrictive happens. The enthalpy change over a complete cycle is zero by definition, and the store is adiabatic (H6), so

$$\eta_{\text{storage}}(\text{CSS}) \equiv 1 \quad (11)$$

identically — for every exchanger size and every flow rate. This is not a result about the store; it is energy conservation applied to a closed cycle in a lossless model. But it has a consequence for how the problem may be posed.

A single-cycle model sizes the field by requiring it to absorb a specified energy within the charging window, and solves for N_{wells} . At cyclic steady state the absorbed and recovered energies are the *same number*, so that requirement and the corresponding discharge requirement collapse into one equation. That equation fixes the ratio of the two flow rates and says nothing whatever about N_{wells} . **There is no root to find.**

So the problem is reposed. Specify the hardware; march to cyclic steady state; report the answer:

quantity	how it is set
N_{wells} $\dot{m}^{\text{ch}}, \dot{m}^{\text{dc}}$	$\Delta E_{\text{out,HP}}/(\rho V_{\text{well}} h_m)$ — latent heat alone each pinned by its own glide
deviation of delivered energy from target	reported

Sizing on latent heat alone *overestimates* N_{wells} , because it ignores the sensible capacity the store also has. That is deliberate and conservative, and the deviation tells you by how much.

The deviation is the glide shortfall. Because the discharge flow is pinned by the *assumed* glide, $\dot{m}^{\text{dc}} = \dot{Q}_{\text{in,ORC}}/(c_p \Delta T_{3C,2C})$, the energy delivered over a fixed window is nothing more than

$$\frac{\Delta E_{\text{delivered}}}{\Delta E_{\text{required}}} = \frac{\text{glide the water actually achieves}}{\Delta T_{3C,2C}}. \quad (12)$$

The two agree to 2×10^{-5} . This is a far more useful statement than “the store is rate limited”, because it is something you could check with two thermometers.

A trap worth avoiding. The deviation cancels out of the round-trip efficiency *exactly*. A +1.2% deviation means 1.2% more electricity out *and* 1.2% more in, because the charge is pinned by its glide in the same way. So the deviation is a statement about **plant size**, not about performance: this field is 1.2% larger than a 1 MW plant, in both directions at once. Computing a round-trip efficiency from the delivered energy and the *target* input would flatter it by that amount.

One correction pass. The ORC evaporating temperature follows the exchanger outlet, which is now a result. So the calculation is run twice: once with a provisional outlet, then again with the value the first pass produced. One pass is enough, and that is a property of the problem rather than an optimistic choice — the outlet is set by the store, not by the plant. Over a sweep in which the inlet moves 9 K and the deviation moves twelve percentage points, the realised outlet moves less than 0.3 K.

9 What a result looks like

Table 1 is the default case. Reproduce it before changing anything: if your first run does not match, something is wrong with the setup rather than with the physics.

Table 1: The design point, at cyclic steady state.

N_{wells}	11.632	from latent heat alone, not solved
$\dot{m}^{\text{ch}}, \dot{m}^{\text{dc}}$	0.966, 0.971	kg/s per leg-pair
cycles to convergence	70	
η_{storage}	1.000000	Eq. (11)
deviation	+4.527	%
implied net output	1.0453	MWe against a target of 1.0000
realised discharge glide	57.44	K against 55 assumed
realised outlet T_{2d}	148.6	°C
ϵ_{local} at end of charge	0.9715	82 of 100 segments reach 1
ϵ_{local} at end of discharge	0.0116	it very nearly refreezes

Performance indicators

Table 2 is the set carried over from the factorial study. These are the numbers to collect across a parametric run, and each answers a different question.

Table 2: Performance indicators at the design point, per cycle.

	meaning	value	what moves it
η_T	discharge thermal $\rightarrow$ net electric	0.1898	the power block alone
$\eta_{T,\text{eff}}$	the same, less discharge pumping	0.1850	tube bore, flow, well depth
η_{RTE}	round trip, both parasitics	0.4367	COP, η_{ORC} , parasitics
ε_{RTE}	$\eta_{\text{RTE}} \times$ cycled fraction	0.4192	anything leaving PCM unused
ΔE_{therm}	thermal energy per well	4734 kWh	$h_m, \rho, V_{\text{well}}$
ΔE_{elec}	net electric energy per well	876 kWh	the above $\times \eta_{T,\text{eff}}$
ρ_E	thermal energy density	171.0 kWh/m ³	h_m, ρ
cycled fraction	$\varepsilon_{\text{ch}} - \varepsilon_{\text{dc}}$	0.9599	
pumping	charge / discharge	26.0 / 26.4 kW	5.02 % of gross

Two things about reading these across a factorial table, both of which are easy to get wrong.

η_{RTE} **does not move with the deviation, at all.** It is the point made at the end of §8: a field delivering 1 % above target also drew 1 % more in, because both flows are pinned by their glides. If a sweep changes the deviation and leaves η_{RTE} where it was, it changed the *size* of the plant and nothing else. Do not report the two as though they were independent findings.

ε_{RTE} **is weighted by the *cycled* fraction**, ε at end of charge minus ε at end of discharge — not by the melted fraction, as in the earlier first-cycle study. The melted fraction *saturates* under this formulation: once a segment has melted all its PCM the further energy goes into superheat, where ε cannot see it. Here 82 of the 100 segments sit at exactly 1 at the end of the charge, so most of the exchanger contributes nothing to the weighting and the rest measures only how far the remaining few fall short. The cycled fraction does not saturate: a store that fills completely and half empties returns 0.5, which is the statement you want.

Profiles

Figure 3 is the picture to carry in your head. Three things are visible in it.

The sawtooth is the cascade. Each layer melts fastest at its leading edge, where the water is hottest relative to *that layer's* melting point, and slowest at its trailing edge. The discontinuities are the layer boundaries. A profile without a sawtooth means your cascade is not doing anything.

The store very nearly fills. By 10 h, 82 of the 100 segments are at $\varepsilon_{\text{local}} = 1$ and the mean is 0.9706. That is the latent-only sizing rule doing its job — and it is also the moment when an area-based model would be discarding heat across most of the exchanger.

And it very nearly empties. The residual after discharge is 0.0116. That residual is the diagnostic that matters most in a parametric study, and §10 explains how to read it. Note what it says here: at this design point the store is closer to *inventory*-limited than to rate-limited.

10 The parameters, and how they are coupled

This is the section to read before your first sweep. Several parameters do more than one job, and the second job is usually the one that surprises you.

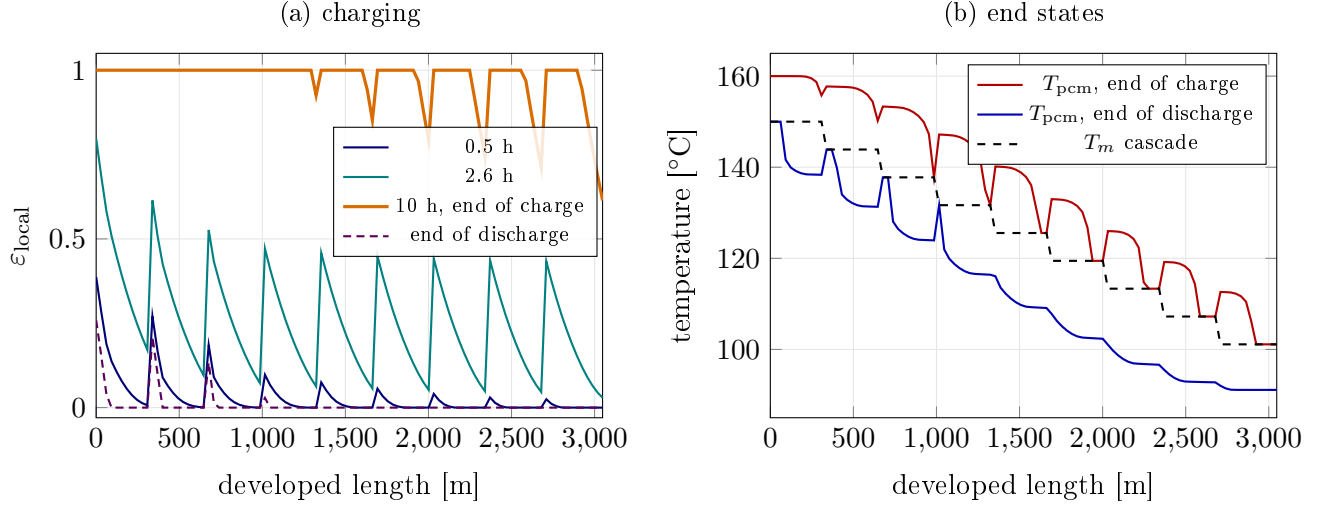


Figure 3: The default case at cyclic steady state. (a) The melted fraction along the tube during charging; the sawtooth is the cascade, each layer melting fastest where the water first meets it. (b) The PCM temperature at the end of each half-cycle against the melting-temperature staircase. Where the solid curve lies above the dashed line the PCM is superheated liquid; below it, subcooled solid. **A model carrying melted area can only ever draw the dashed line.**

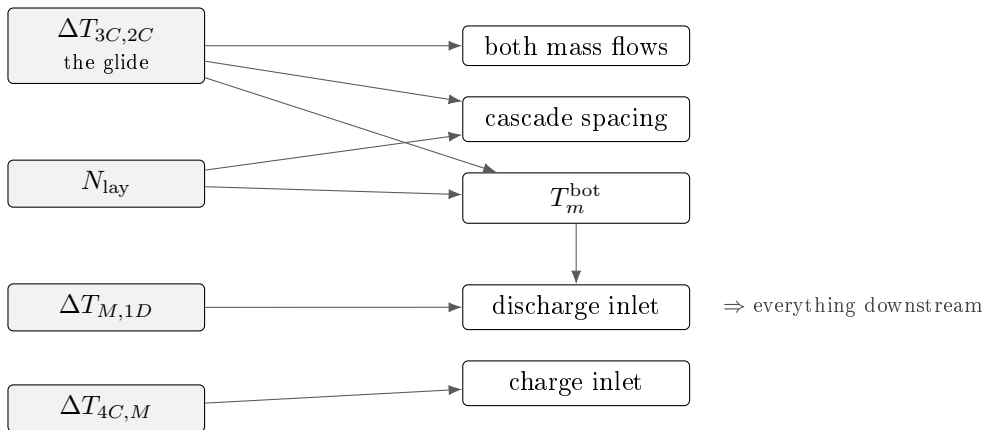


Figure 4: Parameter couplings. $\Delta T_{4C,M}$ and $\Delta T_{M,1D}$ are the two *clean* knobs: each moves exactly one inlet and nothing else. The glide and the layer count each do three jobs.

The trap. Change the glide expecting a mass-flow effect and you have also moved the melting-temperature ladder and, through Eq. (9), the discharge inlet. If you want to vary the flow alone, vary the glide and adjust $\Delta T_{M,1D}$ to hold $T_m^{\text{bot}} - \Delta T_{M,1D}$ fixed.

Rate or inventory? Every parameter acts on one of two things, and telling them apart is most of the skill:

acts on the inventory	acts on the rate	acts on both
h_m, ρ	k_s, k_l	$\Delta T_{3C,2C}$
$V_{\text{well}}, \text{well depth}$	fins (n_f, L_f, t_f)	N_{lay}
	tube radii	T_m
	$\Delta T_{4C,M}, \Delta T_{M,1D}$	
	window lengths	

The residual melted fraction at the end of discharge is the instrument. If it is near zero, the store empties completely and you are *inventory* limited: more PCM would help. If it is well above zero, and particularly if it *rises* when you add wells, you are *rate* limited: more PCM will not help, and the lever is flow, window length or conductance.

A sweep that moves the deviation a lot and the residual hardly at all is changing the size of the plant. One that moves the residual is changing its physics.

Do not tune to zero deviation. It is tempting, when a sweep crosses the 1 MW target, to adopt that parameter value. Resist it. The deviation is the one honestly reported output of the whole procedure; choosing a parameter so as to zero it converts it back into a satisfied constraint and throws away the information. Choose parameters for reasons — $\Delta T_{M,1D} = 10 \text{ K}$ is used because it is symmetric with the charging superheat — and let the deviation say what it says.

11 Running it

In the notebook, §3.1 is the only cell you need to edit. **Case** is a frozen dataclass, so variants are made by copying:

```
case = CASE.with_(h_m = 180_000.0,      # J/kg, not kJ/kg
                  N_lay = 12)
validate_case(case)
css = simulate_css_corrected(case, record=True, n_cycles=N_CYCLES)
css_report(case, css)
```

Four things that will otherwise cost you an afternoon:

1. **Units are SI base units.** h_m is in J/kg, so 180 kJ/kg is 180_000.0. Writing 180 gives a store with essentially no capacity, and no error.
2. **Edit §3.1, not the model cell.** Rewriting a default inside the `%writefile` cell changes the file on disk but not the module already imported, so the old value is used and the old answer is reported without complaint.
3. **Watch the convergence banner.** `N_CYCLES` sits in §3.1 directly under the case, because it is the one *numerical* setting you will have to change. It is a give-up limit on the cyclic-steady-state loop, and cycles to convergence grow with the exchanger count and

with anything that makes each unit work less hard: the design point needs about 70, $N_{\text{wells}} = 16$ needs well over 200, and $h_m = 180 \text{ kJ/kg}$ can need 300 or more.

The failure mode is what makes this worth a paragraph. A run that has not converged does not crash and does not return NaN: it prints a *complete* set of results, indistinguishable from a good one except for the ***** NOT CONVERGED ***** banner above them. The store had not returned to its own state, so the deviation, the melted fractions and every index below that line mean nothing. In a sweep the banner can scroll past inside a loop, so **sweep** prints one line per offending point and carries a **converged** column in the table it returns — check it before plotting anything.

4. **Run validate_case every time.** It does not check that the model is right — that is settled elsewhere — but that *your parameters* are consistent: layer boundaries falling inside segments, an inlet on the wrong side of the cascade, an approach narrower than one layer width.

12 What the model does not contain

A parametric study can wander into the region where one of these dominates, so they are worth knowing by name.

1. **No ground heat loss (H6).** The store is adiabatic, so $\eta_{\text{storage}} \equiv 1$ at cyclic steady state *by construction*. No sweep can produce a storage loss, and you should not read the perfect round trip as a result.
2. **H3 is saturated.** At the design point the melt fronts of neighbouring tubes are in contact over 99 % of the exchanger, so the annular conduction resistance is being evaluated at the exact limit of its validity. Sweeps that *increase* melting — more fins, higher k , longer charging — go further into that region, not out of it.
3. **H9 is unquantified and not small.** The two legs of a hairpin are up to 48.9 K apart at the wellhead and are assumed perfectly insulated from one another. The neglected leak is of order 25 % of the useful duty.
4. **No validation against experiment** or against a higher-fidelity model. Energy closure is structural: it shows the implementation is self-consistent, not that it is right. This is the largest open item.
5. **Idealised cycles.** The compressions and expansions carry no isentropic efficiency, so the COP and η_{ORC} are optimistic.
6. **No volume change on melting**, and the melt-front shape around the fins is not modelled.

A deviation of a few per cent sits *inside* the uncertainty of items 1–3. Use it to compare cases, not as an absolute.

Where the rest is

P2H2P_model.ipynb	this document, executable
P2H2P_verification.ipynb	fixtures, regression checks, version history
DESIGN_NOTES.md	why each superseded approach was superseded
project report	the full model statement, equation by equation
manuscript	the formulation and its numerics, for publication